Supplement to "Learned representations and kernel corrections for radiative-transfer emulation"

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Abstract

This supplement contains the proofs of the results stated in the article, three further results on the sensitivity, selection and perturbation of learned corrections, and the experiments that support the article without being needed to follow it: further representation and combination studies on the EMIT table, a synthetic check of the transfer theorem, a matched correction-coefficient benchmark, full-data tuning and gradient-based metrics, a comparison on OCO-2, ten further emulation configurations and extended tables. Sections, results, tables and equations with the prefix S are in this document; numbers without it refer to the article.

S1 Sensitivity, selection and perturbation of learned corrections

S1.1 Sensitivity of the correction

For a common frozen operator $R = I - S_\lambda$ and the norm of Section 8, the exact criterion $2\langle Rf, Rh\rangle > \|Rh\|^2$ is necessary and sufficient for strict improvement of the corrected predictor over the standalone kernel, since $\|Rf - Rh\|^2 < \|Rf\|^2$ is that inequality rearranged; the comparison of smoothness norms is neither necessary nor sufficient. Write the design values of the targets, of the fitted mean and of the corrected predictor as y, h_X and $\widehat{y}$, with $\widehat{y} = h_X + S_\lambda(y - h_X)$ and $S_\lambda = K(K + \lambda I)^{-1}$; in the nugget convention of the experiments, $\lambda = n\gamma$. The mean is trained on the same labels, so $h_X = h_X(y)$. In the following Jacobian calculation, K and λ are held fixed; label-dependent feature or kernel selection would add terms involving derivatives of S_λ.

Proposition S1.1. *1. If h_X is differentiable in y, the Jacobian of this fixed-kernel pipeline is $J_{\widehat{y}} = S_\lambda + (I - S_\lambda)J_h$. In particular the correction stage alone does not bound the sensitivity of the pipeline: with $K = I$, $\lambda = 1$ and $h_X(y) = 2y$ one has $\widehat{y} = \frac{3}{2}y$ although $\|S_\lambda\|_{\mathrm{op}} = \frac{1}{2}$; and if the mean interpolates the training labels, $h_X(y) = y$, then $\widehat{y} = y$ and the correction changes nothing on the design.*

2. With the mean and the kernel frozen, $y \mapsto \widehat{y}$ is affine with linear part S_λ, its effective degrees of freedom are $\operatorname{tr} S_\lambda = \sum_j \mu_j/(\mu_j + \lambda)$, and $\|S_\lambda\|_{\mathrm{op}} = \max_j \mu_j/(\mu_j + \lambda) < 1$.

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3. *Suppose the mean, the kernel and the ridge are fixed independently of the correction noise, conditional on the design, and the correction is fitted on an independent sample with* $y = f_X + \varepsilon$, $\mathbb{E}\varepsilon = 0$, $\mathbb{E}[\varepsilon\varepsilon^\top] = \sigma^2 I$. *With* $r = f_X - h_X$ *and* r_j *its coordinates in the eigenbasis of* K,

$$\frac{1}{n}\,\mathbb{E}_\varepsilon\|\widehat{y} - f_X\|^2 = \frac{1}{n}\sum_j \Big(\frac{\lambda}{\mu_j + \lambda}\Big)^2 r_j^2 + \frac{\sigma^2}{n}\sum_j \Big(\frac{\mu_j}{\mu_j + \lambda}\Big)^2.$$

4. *At a new input* x, *with* $a_\lambda(x) = (K + \lambda I)^{-1}k(X_n, x)$, *the noise variance of the correction is* $\sigma^2\|a_\lambda(x)\|_2^2$.

Proof in Section S9. Item (iii) is what the ridge-path experiments report: the residual's spectral energy $\sum_j r_j^2$ by eigendirection, and the two sums as functions of λ, together with the out-of-sample factor of item (iv). A ridge chosen from fits that depend on the correction noise is outside the hypothesis of (iii).

S1.2 A finite-class selection guarantee

A finite comparison on a genuinely independent selection sample has a simple risk bound [11, 12, 13]. The independence assumption is stronger than reusing the same validation rows for training decisions and combination weights.

Proposition S1.2. *Let* $p_1, \dots, p_M$ *be predictors frozen before an independent selection sample of* m *i.i.d. units, let the loss take values in* $[0, B]$, *and let* $\widehat{p}$ *minimise the empirical risk on that sample. Then with probability at least* $1 - \delta$,

$$R(\widehat{p}) \le \min_{1\le j\le M} R(p_j) + 2B\sqrt{\frac{\log(2M/\delta)}{2m}}.$$

Proof in Section S9. The squared constrained-retrieval loss is bounded by R^2. Thus the proposition can compare a finite set of frozen corrections or stacks, with bands and prescribed reflectances averaged within each independent state. For example, a finite grid of damped corrections $h + \alpha k(x, X_n)(K + \lambda I)^{-1}(y - h_X)$ can include $\alpha = 0$; the standalone kernel must be added as a separate candidate when it is not already in the grid. To apply the result to stacks, one first fits their weights on development rows and then compares a frozen finite collection on an independent state-level sample. In the experiments of the article the same validation rows serve early stopping, kernel tuning and combination, so the proposition describes the protocol under which such a comparison is certified rather than a property of the reported stacks.

S1.3 Perturbing a learned-mean kernel predictor

Fix the training sample and write a corrected predictor as $p(x) = h(x) + k_x^\top H^{-1}(Y - M)$ with $H = K + \lambda I$ positive definite, $Y \in \mathbb{R}^{n\times q}$ the targets and $M \in \mathbb{R}^{n\times q}$ the training-row means used as correction labels. Let a second fitted system be $\widetilde{p}(x) = \widetilde{h}(x) + \widetilde{k}_x^\top \widetilde{H}^{-1}(Y - \widetilde{M})$ with $\widetilde{H}$ symmetric positive definite; both systems may depend on Y through the mean, the features, the selected ridge and any kernel approximation. Put $B = H^{-1/2}(\widetilde{H} - H)H^{-1/2}$, whose least eigenvalue exceeds -1, $Q = (I + B)^{-1}$, $\gamma_Q = \|Q\|_{\mathrm{op}} = 1/(1 + \lambda_{\min}(B))$, $\eta_Q = \|Q - I\|_{\mathrm{op}} = \max_i |b_i|/(1 + b_i)$ over the eigenvalues b_i of B, and

$$g = H^{-1/2}k_x, \quad \widetilde{g} = H^{-1/2}\widetilde{k}_x, \quad v = \widetilde{g} - g, \quad W = H^{-1/2}(Y - M), \quad U = H^{-1/2}(\widetilde{M} - M).$$

A conditioning argument bounds such a difference by the condition number of H times a norm of the change [26, 24, 27]. The statement below instead keeps the residual action W and the label shift U separate, which is what makes it readable on data.

Proposition S1.3 (same-data perturbation)**.** *1. The exact difference is*

$$\widetilde{p}(x) - p(x) = \widetilde{h}(x) - h(x) + v^\top W - (Q\widetilde{g})^\top (BW + U).$$

2. $\|\widetilde{p}(x) - p(x)\|_2 \le \|\widetilde{h}(x) - h(x)\|_2 + \|W^\top v\|_2 + \|Q\widetilde{g}\|_2 \, \|BW + U\|_{\mathrm{op}}$.

3. $\|\widetilde{p}(x) - p(x)\|_2 \le \|\widetilde{h}(x) - h(x)\|_2 + \|W\|_F\big(\gamma_Q\|v\|_2 + \eta_Q\|g\|_2\big) + \gamma_Q\|U\|_F\|\widetilde{g}\|_2$.

4. If $\widetilde{H} = H + sI$ *with* $s \ge 0$ *and the kernel and the means unchanged, then* $\|\widetilde{p}(x) - p(x)\|_2 \le \|g\|_2\|W\|_F \, s/(\lambda_{\min}(H) + s)$.

Proof in Section S9. Only positive definiteness of $\widetilde{H}$ is needed. A large positive eigenvalue of B, as can arise from increasing the ridge, does not invalidate the bound. The bound can nonetheless be numerically large, especially when the least eigenvalue approaches -1. Item (ii) keeps the interaction between the perturbation and the residual: with $v = 0$, $U = 0$ and $BW = 0$ the kernel correction does not move, however large $\|B\|$ is; the full predictor is unchanged as well when $\widetilde{h}(x) = h(x)$. The bounds are stated in the residual W, not in Y, they use no independence between the representation, the mean, the ridge and Y, and they transfer between fitted predictors without being generalisation statements.

For component j the physical target at band b is $f_{j,b} = \mu_{j,b} + \sum_\ell (A_j)_{b\ell} z_{j,\ell} + \pi_{j,b}$, with A_j the standardiser scale times the retained PCA basis and π_j the truncation residual, and the pipeline predicts the retained coordinates. For a frozen realised system $p_\ell(x) = h_\ell(x) + a_x^\top(y_\ell - h_{\ell,X})$ with $y_\ell = f_{\ell,X} + \varepsilon_\ell$, the conditional RKHS evaluation bound [3, 4] gives

$$|p_\ell(x) - f_\ell(x)| \le B_\ell \widetilde{P}_\lambda(x) + |a_x^\top \varepsilon_\ell|, \tag{S1.1}$$

where, writing $k_x = k(X_n, x)$ and $a_x = (K + \lambda I)^{-1} k_x$,

$$\widetilde{P}_\lambda(x)^2 = k(x,x) - 2a_x^\top k_x + a_x^\top K a_x = k(x,x) - k_x^\top (K + \lambda I)^{-1} k_x - \lambda\|a_x\|_2^2.$$

This follows by applying the RKHS Cauchy–Schwarz inequality to the residual and $k(x,\cdot) - \sum_i (a_x)_i k(x_i,\cdot)$. Here B_ℓ is a bound on the residual RKHS norm, a hypothesis and not an observed certificate, and the second term is zero for deterministic simulator targets, which is the form used for EMIT. A perturbed system adds the bound $\Delta_\ell(x)$ of Proposition S1.3. If $|\widehat{z}_{j,\ell} - z_{j,\ell}| \le u_{j,\ell}$, then

$$|\widehat{f}_{j,b} - f_{j,b}| \le |\pi_{j,b}| + \sum_\ell |(A_j)_{b\ell}| \, u_{j,\ell}, \tag{S1.2}$$

and a change of basis or standardiser between two systems adds its own reconstruction term. Assembling $e_R = |e_a| + R|e_t| + R^2 t|e_s|$ from (S1.2) for the three physical components and applying Theorem 4 closes the chain. If the inverse is evaluated at $L_{\mathrm{obs}} = L + \eta$, the proof of Theorem 4 goes through with e_a replaced by $e_a - \eta$, so $|\bar{\rho} - \rho| \le \min\{R, (|e_a - \eta| + R|e_t| + R^2 t|e_s|)/t\}$. The theorem is an exact-arithmetic statement. In floating point the added term $\rho t/q$ can be lost when it is small relative to the spacing of representable numbers near a, in a way that depends on ρ, q, the magnitudes and the rounding rather than on a single threshold such as $t < 2^{-52}a$; a rounded radiance may then give zero for $L - a$ even with the tabulated components. A count of numerical violations is therefore a check on the implementation, not a test of the inequality.

S2 Representation, correction and combination

Concatenated features. Table S1 gives the three-member width-2000 experiment of Section 7 split by split, with the networks trained on one GPU.

The concatenated-feature kernel has mean radiance error 0.0688%, against 0.0764% for the features of one network; the paired gain is 0.0076 ± 0.0053 percentage points, positive on three splits and -0.0002 on the fourth. Its mean within-split 95th-percentile reflectance error is 5.99 points, against 7.51 for single-member features and 14.35 for the stack. With the ten width-512 splits of Section 7 this makes two settings, with different networks on overlapping splits, in which concatenation lowers both the radiance error and the reflectance tail.

Kernel choice on the state and on learned features. Table S2 covers four components on seeds 101–103. The spectral mixture and the Matérn-5/2 kernel have similar mean errors on Y_1 (0.089%), and the spectral mixture has the lowest mean errors of the raw-state kernels on Y_2 and Y_3 (0.237% and 0.099%). The first-order additive kernel is much less accurate in this specification. Albedo errors for the nonadditive kernels lie between about 2.3% and 3.5%.

The Matérn-5/2 feature kernel gives 0.105%, 0.211%, 0.114% and 2.303% on Y_1–Y_4, better than the best raw-state kernel on some components and not on others. The convex stack has the lowest component means in this comparison (0.066%, 0.156%, 0.077%, 1.956%), which is consistent with its large reflectance tails elsewhere: the component metric does not weight errors the way the inverse does.

Network regularization. Table S3 compares a control, a weight-decay grid and an early-stopping grid on three seeds. The validation-selected weight decay matches the control's component mean (2.27%), and the selected early stopping is less accurate and more variable (2.72 ± 0.45%). Residual correction brings the component means to 1.88–1.95% and feature kernels to 0.95–0.99%, with radiance errors near 0.15%, whichever regularization the network used. The reference row is the isotropic kernel fitted in the same runs. Under this budget, regularizing the network changes little; the kernel stage does the work.

Perturbation and ridge-path diagnostics. The tables of Section S8 give the perturbation diagnostics for the state inputs and the learned features. The residual-action bound of Proposition S1.3 is often much smaller than the spectrum-only bound, yet it can still exceed the actual movement by a factor of tens, thousands or more: positive definiteness makes the bound applicable without making it a useful tolerance.

Table S8 reports a 4,000-row fit of four components at seed 101. At the smallest ridge on input-space Y_1 the effective degrees of freedom are 3,841 and the noise factor 0.924. The same-half and split-half principal-component errors are 0.079% and 2.465%, and the physical-unit test errors of the corrected and standalone kernels 0.2168% and 0.1754%; these columns use different metrics and samples and are not to be subtracted. At $\gamma = 10^{-2}$ the corresponding test errors are 0.5095% and 5.7578%: the fitted mean keeps its predictive information when the kernel correction is strongly regularized.

seed	head	radiance [%]	median [pp]	p95 [pp]	gain [pp]
104	feature kernel	0.0924	0.044	9.25	–
104	concatenated features	0.0810	0.025	6.25	+0.0114
104	convex stack	0.0781	0.024	12.77	–
104	network + residual	0.1843	0.115	54.71	–
105	feature kernel	0.0611	0.031	7.31	–
105	concatenated features	0.0508	0.028	6.02	+0.0103
105	convex stack	0.0475	0.024	14.50	–
105	network + residual	0.1349	0.087	42.61	–
106	feature kernel	0.0789	0.043	7.67	–
106	concatenated features	0.0702	0.035	6.28	+0.0087
106	convex stack	0.0687	0.034	16.29	–
106	network + residual	0.1608	0.108	51.84	–
107	feature kernel	0.0731	0.042	5.82	–
107	concatenated features	0.0732	0.044	5.41	-0.0002
107	convex stack	0.0833	0.043	13.84	–
107	network + residual	0.1476	0.103	46.58	–
mean	feature kernel	0.0764±0.0130	0.040±0.006	7.51±1.41	–
mean	concatenated features	0.0688±0.0128	0.033±0.009	5.99±0.40	0.0076±0.0053
mean	convex stack	0.0694±0.0158	0.031±0.009	14.35±1.48	–
mean	network + residual	0.1569±0.0211	0.103±0.012	48.93±5.40	–

Table S1: Kernel on concatenated features, three width-2000 networks, splits 104–107. Radiance is a mean relative spectral error in percent; reflectance median and 95th-percentile errors are in percentage points. The last column is the paired radiance gain of concatenated features over the features of one network. The mean row averages the four within-split quantities.

head	Y_1 [%]	Y_2 [%]	Y_3 [%]	Y_4 [%]
Matérn-1/2	0.179±0.008	0.672±0.003	0.418±0.012	2.326±0.103
Matérn-3/2	0.092±0.002	0.391±0.008	0.207±0.016	2.481±0.043
Matérn-5/2	0.089±0.017	0.434±0.007	0.203±0.003	2.746±0.082
Gaussian	0.092±0.002	0.525±0.113	0.228±0.051	3.502±0.501
spectral mixture (4)	0.089±0.010	0.237±0.014	0.099±0.013	3.468±0.263
NNGP, ReLU	0.175±0.001	0.674±0.015	0.350±0.009	2.548±0.057
NTK, ReLU	0.263±0.002	1.055±0.015	0.621±0.014	2.321±0.048
additive Matérn-5/2	4.091±0.093	20.266±1.035	11.430±0.237	4.413±0.077
ARD by kernel flow	0.322±0.015	0.257±0.019	0.199±0.005	2.418±0.114
ARD by marginal likelihood	0.281±0.039	0.341±0.042	0.183±0.011	2.702±0.027
convex kernel sum	0.087±0.007	0.253±0.020	0.099±0.013	2.557±0.099
network	0.819±0.027	2.289±0.019	0.910±0.057	3.482±0.085
network + residual kernel	0.523±0.007	1.337±0.026	0.588±0.035	3.055±0.121
features, Matérn-5/2	0.105±0.006	0.211±0.016	0.114±0.008	2.303±0.216
features, Gaussian	0.126±0.007	0.262±0.018	0.138±0.014	2.426±0.214
features, NTK	0.152±0.005	0.307±0.015	0.156±0.007	2.150±0.059
features, ARD	0.116±0.007	0.214±0.014	0.111±0.004	2.217±0.207
coordinate selection	0.090±0.010	0.360±0.008	0.105±0.010	2.797±0.447
convex stack of the heads	0.066±0.003	0.156±0.010	0.077±0.004	1.956±0.037
ridge stack of the heads	0.087±0.005	0.288±0.031	0.087±0.005	2.310±0.071

Table S2: Ten input-space kernels, feature kernels and combination rules on the EMIT state inputs, with rank-64 outputs. Entries are component mean relative spectral errors in percent, averaged over seeds 101–103 with sample standard deviations. The network has width 512 and depth 3 with a 150-epoch budget. Kernels include the Matérn family, Gaussian, spectral mixture [9], NNGP [17, 16], NTK [18], additive kernel [10], and learned input scales [7, 3].

head	alone component / radiance	residual correction component / radiance	feature kernel component / radiance
	0.941±0.109		
isotropic kernel	0.284±0.078	–	–
	2.266±0.031	1.903±0.055	0.955±0.038
network, control	0.809±0.029	0.617±0.033	0.153±0.052
	2.269±0.027	1.880±0.072	0.948±0.048
network, weight decay	0.820±0.014	0.618±0.040	0.154±0.048
	2.716±0.445	1.949±0.073	0.986±0.032
network, early stopping	1.293±0.373	0.789±0.127	0.147±0.054

Table S3: Regularization comparison, seeds 101–103. Each cell gives the mean component relative error [percent] / radiance relative error [percent], with sample standard deviations. The width-512, depth-3 network has a 150-epoch cap, and validation selects among three runs per arm. The weight-decay grid is $10^{-5}, 10^{-4}, 10^{-3}$ and the early-stopping patience grid 3, 5 and 10 epochs [14, 15]. The reference row is the isotropic kernel fitted in the same runs.

S3 A synthetic check of the transfer theorem

Table S9 checks the construction of Proposition 5 on manufactured data, two million samples for each $\beta \in \{0.5, 1, 2, 3, 4\}$. The average-error slopes are 0.200, 0.333, 0.498, 0.597 and 0.664, against the limits 0.200, 0.333, 0.500, 0.600 and 0.667. For the uniform family the finite-ladder slope is compared with the analytic curve on the same ladder: at $\beta = 2$ the measured slope is 1.721 and the analytic one 1.724, while the limit is two. The largest relative Monte Carlo discrepancy is about 1.01% for the uniform family at $\beta = 2$ and below 0.053% for the profile family against its quadrature reference, and no pointwise violation occurs among the ten million sampled transmissions.

S4 The matched benchmark and the transfer contrast

Correction-coefficient benchmark. Table S10 scores our families under the standard and out-of-distribution protocols of [2], with and without the low-fidelity 6S coefficients as inputs, against the published values of the reference model, which we did not retrain.

The state-only residual-corrected network reaches standard-split RMSE 0.00581 ± 0.00004, MAE 0.00151, uniform-average $R^2 = 0.9952$ and SMAPE 3.54%, against published values 0.00619, 0.00186, 0.9947 and 3.91%; with 6S inputs its means are 0.00578, 0.00148, 0.9954 and 3.19%. On the out-of-distribution split the state-only means are 0.00620, 0.00252, 0.9967 and 3.37%, against published 0.01022, 0.00544, 0.9942 and 5.40%, and with 6S inputs 0.00596, 0.00232, 0.9969 and 2.92%. The spreads describe three seeds on the standard protocol and three initializations on the out-of-distribution protocol.

The R^2 convention matters here. The reference weights the three coefficients equally, whereas a variance-weighted R^2 is the default of many libraries. We report the uniform average, computed from per-coefficient target scales that are recovered from the per-coefficient RMSE of the heads and checked against the variance-weighted value. The ensemble without correction is close to the corrected network, while the raw-state kernel is much less accurate in these configurations.

OCO-2 coefficient and radiance objectives. Table S11 covers three bands and seeds 0–2. The feature kernel trained under the flat coefficient loss has coefficient errors of about 4.1%,

	component rel. L^2 [%]				radiance	reflectance
kernel	Y_1	Y_2	Y_3	Y_4	rel. L^2 [%]	med. $\lvert\hat{\rho}-\rho\rvert$ [%]
isotropic, tuned on 6,000 rows	0.08±0.01	0.42±0.03	0.20±0.00	2.62±0.18	0.239±0.016	0.141±0.018
isotropic, tuned on every row	0.08±0.00	0.36±0.02	0.17±0.01	2.58±0.14	0.210±0.008	0.107±0.006
per-input search, tuned on 6,000 rows	0.08±0.01	0.16±0.02	0.08±0.01	2.18±0.16	0.095±0.009	0.036±0.004
per-input search, re-tuned on every row	0.08±0.00	0.16±0.02	0.08±0.01	2.09±0.12	0.096±0.009	0.036±0.004
per-input from the cubic's gradients, $p = 1$	0.34±0.01	0.19±0.02	0.14±0.01	2.17±0.14	0.159±0.007	0.078±0.004
per-input from the cubic's gradients, $p = 1/2$	0.11±0.00	0.19±0.02	0.09±0.01	2.27±0.15	0.125±0.008	0.055±0.004
per-input from the cubic's gradients, $p = 1/4$	0.08±0.00	0.25±0.02	0.14±0.01	2.38±0.20	0.157±0.015	0.077±0.013

Table S4: Where the kernel's hyperparameters are chosen, and how its metric is obtained, at the full training block and the same ten splits. The first pair tunes smoothness, scale and nugget on a 6,000-row subsample and on every training row; the last three rows replace the coordinate search by a metric read off the cubic surrogate's gradients, at three exponents. Selecting the exponent on validation, per split, chooses $p = 1/2$ on all ten, so that row would repeat the $p = 1/2$ row and is omitted. Mean ± standard deviation over the ten splits.

16.3% and 8.1% on O_2, weak CO_2 and strong CO_2, and radiance errors of 0.076%, 0.114% and 0.108%. A radiance-weighted feature construction gives about 0.031%, 0.048% and 0.064% in radiance, at the price of larger coefficient errors. The reconstruction map and the evaluation metric decide which fitted representation is preferred.

The radiance stack minimizes a row-relative squared radiance loss, a convex quadratic because the reconstruction is affine, and its solver reaches KKT gaps between 4.17×10^{-6} and 1.52×10^{-5}. Its mean radiance errors are 0.0264%, 0.0391% and 0.0511%, against 0.0291%, 0.0404% and 0.0536% for the squared-loss stack on reduced coordinates, while its coefficient errors rise to 18.59% and 37.43% on O_2 and weak CO_2 and to 8.87% on strong CO_2. All rules use 2,000 validation and 2,000 test rows.

S5 Full-data tuning and gradient-based metrics

We compare hyperparameter selection on a 6,000-row subsample with selection on all training rows, and coordinate search with a metric read off the gradients of a cubic surrogate (Table S4).

Tuning with every row in the solve. The kernel families of Section 4 choose smoothness, scale and nugget by the validation error of a solve on a 6,000-row subsample and refit the winner on all 18,884 rows. Tuning with the full solve instead lowers the isotropic kernel's radiance error from 0.239% to 0.210% and its median reflectance error from 0.141 to 0.107 points, better on nine of ten splits, with the ratio averaging 0.882 and never above 1.001. The selected scales differ: over the forty component-split cells the full block picks a shorter length scale than the subsample in 26, the same in 12 and a longer one in 2. A smaller tuning fit prefers more smoothing.

Below 6,000 rows the tuning subsample is the whole training block, so those rungs are already fully tuned; at 8,000 rows it is three quarters of the block and only at the top rung is it a third. Applying the measured ratio to the top rung of Table 10 takes the kernel's last step on radiance from -0.183 to -0.334, against a slope of -0.305 over the five lower rungs, and on the two flux components from -0.187 and -0.171 to -0.352 and -0.358, against -0.437 and -0.468. Most of the bend of Section 7 is the tuning protocol, and with full tuning the kernel's margin over the network at the full block is 1.8 rather than 1.6.

Full-data tuning barely changes the input-scaled kernel: 0.096% against 0.095%, better on four of ten splits, ratio 1.005. Its coordinate search already arrives at the scale the full block selects.

Gradient-based metric. The coordinate search costs a sweep over the six inputs at five multipliers, twice, per component and split. The alternative fixes the metric before any kernel is fitted: the multiplier of coordinate j is the root-mean-square gradient of the fitted cubic surrogate along j over the training rows, raised to a power p, normalized to geometric mean one and clipped to $[1/16, 4]$, the range of the multipliers the search selected, so that weakly varying coordinates receive longer length scales. It costs one cubic fit and six finite differences.

At $p = 1$ the radiance error is 0.159%, below the isotropic kernel on every split, by a factor 0.669 on average, and above the searched metric on every split, by a factor 1.680. The dependence on p is not monotone: $p = 1/2$ gives 0.125% and $p = 1/4$ gives 0.157%. Choosing p like every other hyperparameter here, by the lowest mean validation error over the four components, selects $p = 1/2$ on all ten splits. At that exponent the gradient metric closes 79% of the gap between the isotropic and searched kernels on radiance and 82% on the median reflectance error, for one cubic fit and a choice among three numbers.

The components prefer different exponents: Y_2 and Y_3 are best at 1/2, the albedo at 1 (2.17% against 2.27%), and the path radiance at 1/4. On path radiance the isotropic kernel already reads 0.084% and the searched metric 0.085%, while the proportional metric reads 0.34%, four times the isotropic error. The gradient of the surrogate along a coordinate measures how fast the map moves with it, not how much that coordinate helps predict, and on this component the two disagree. Softening the exponent to 1/4 brings Y_1 back to 0.085%, at the cost of Y_2 and Y_3. Coordinate search can adjust each component separately, whereas the gradient construction uses one exponent for all four.

S6 Comparison with OCO-2

The companion paper [4] applies the same construction to the OCO-2 radiative-transfer emulation problem of Lamminpää et al. [1]: per spectral band (O_2, weak CO_2, strong CO_2), a reduced atmospheric state of twenty or twenty-four coordinates maps to forty reduced radiance coefficients, with errors on the standardized coefficients and on the reconstructed radiance, under the ten-split protocol of Section 3. It gives a comparison at a higher input dimension.

There a kernel on the state trails the network by a factor of nine, four and five on the reduced coefficients: the isotropic Matérn reaches 40.6%, 58.6% and 43.4% on the three bands against 4.4%, 16.4% and 8.2% for the network. Four ways of learning the kernel's metric on the state were run on the same ten splits on all three bands: a kernel-flow metric [7], a Mahalanobis metric under the same flow, the gradient outer product of the recursive feature machine [19], and per-input scales fitted by empirical Bayes. The last is the best of the four on every band, taking the isotropic kernel's 40.6%, 58.6% and 44.3% on the coefficients to 18.7%, 25.3% and 16.9%, and on O_2 its radiance from 0.233% to 0.150%; the two flow metrics agree within half a point of each other at 26.9%, 29.8% and 23.9%, the recursive feature machine reads 28.2%, 47.9% and 31.3%, and an additive kernel 38.2%, 46.9% and 47.7%. (The isotropic kernel on strong CO_2 reads 44.3% here and 43.4% in the baseline experiment. The metric experiment also chooses the Matérn smoothness on validation and takes $\nu = 3/2$ on this band at every split; the baseline fixes $\nu = 5/2$, and its choice of scale and nugget changes from split to split, six splits landing near

41%, three near 48% and one at 44%, hence its spread of ±3.4.) The feature kernel remains more accurate, by a margin that depends on the band: the exact Matérn regression on the network's frozen features reads 4.1%, 16.3% and 8.1% on the same splits, so the best state metric is 4.6 times that on O_2 but only 1.6 and 2.1 times it on the two CO_2 bands. The empirical-Bayes metric is bimodal across splits on O_2, eight of ten landing near 17.2% and two at 24.7%, so its spread of ±3.2 is thirty times the isotropic kernel's ±0.1. The exact Matérn regression on the network's frozen features beats a ridge readout refitted on the same features on every split, every band and every training loss, and a per-coefficient choice among heads trained under different losses has lower mean error than the release's own emulator under both reported metrics: it wins the reduced-coefficient comparison on all ten splits on each band, and the radiance comparison on ten, nine and ten splits on O_2, weak CO_2 and strong CO_2.

S7 Ten further emulation configurations

Table S5 extends the comparison to ten further configurations, each with validation-based selection; the comparison concerns the families under a common protocol, not the best method available for each problem.

The corpora are the advection equation with discontinuous initial data from the operator suite of [20], which is also the source of the structural-mechanics problem of [4]; Burgers' equation at three viscosities from PDEBench [21]; three one-step maps from The Well [22]: a two-dimensional turbulent radiative layer, an active-matter system and a Helmholtz staircase; the ClimSim atmospheric physics emulation task [23]; and the paired correction-coefficient corpus of the physics-guided radiative-transfer work of [2], a third radiative-transfer emulation problem and the largest here, with about 488,000 training rows. That corpus is run in both of its configurations: from the atmospheric state alone, and with the low-fidelity 6S coefficients as additional inputs, which is a different and easier problem.

Advection is the 200-point input/output pair of the operator suite at full resolution. Burgers is PDEBench's `1D_Burgers_Sols_Nu`ν release, the map from the initial condition to the solution at $t = 1$ on a grid subsampled by four to 256 cells, with the input used raw and the output reduced to 128 principal coefficients. The three Well datasets are one-step maps of the full multi-channel state with 256 principal coefficients in and out: the turbulent radiative layer on its 128×384 grid (density, pressure, velocity), active matter on 256×256 (concentration, velocity and the two tensor fields), and the Helmholtz staircase on its two pressure channels. ClimSim is the low-resolution release, 124 inputs to 128 tendencies, with 100,000 training rows drawn from the 10.1 million available and validation and test taken from the release's own validation and scoring files. The correction-coefficient corpus is the paired 6S/libRadtran Sentinel-2 set [5, 6]: wavelength, solar zenith, view zenith, relative azimuth, aerosol optical depth, water vapor, ozone and elevation map to three libRadtran coefficients, split by atmospheric state so that all bands of a state stay together; the multi-fidelity configuration adds the three 6S coefficients, for input dimensions eight and eleven. For these two configurations the error is

$$E_{0.05} = \frac{1}{N} \sum_{i=1}^{N} \frac{\|Y_i - \widehat{Y}_i\|_2}{\max\{\|Y_i\|_2, 0.05\}}, \tag{S7.1}$$

and the same floor enters validation selection and the kernel-flow objective, which limits the weight of coefficient vectors of small norm.

The families are those of Section 4 with two changes forced by scale. The reference is a ridge regression linear in the inputs, since a cubic expansion of a 256-dimensional input is not tractable. Kernel solves are exact when the training block has at most 20,000 rows and use a Nyström approximation [8] on 6,000 uniformly drawn landmarks above that, which applies to ClimSim, the Helmholtz staircase at 20,384 rows and the two correction-coefficient configurations. The convex stack and the per-coordinate selection are formed from the kernels, the networks, the residual correction and the feature kernel; the ridge is a reference row, not one of their members. Advection, Burgers, ClimSim and the two correction-coefficient configurations have three to five random splits. The Well's releases carry their own training, validation and test files, which are used unchanged, so the three turbulent-radiative-layer runs share one split and differ only in network initialization (their spread, of order 0.01 points, measures initialization), and the active-matter and Helmholtz-staircase rows are single runs.

The relative errors vary widely with the problem. Where the map is smooth and the sample small, Burgers at $\nu = 0.1$, the exact kernel is at 2.81% against the network's 20.34%, a factor of seven, of the same kind as the five- to six-fold gap between the per-input kernel and the network on the EMIT table's smooth components; as the viscosity falls and the solutions roughen the gap closes to 23.40% against 36.65% and then to 31.66% against 39.98%. Where the sample is large and the map rough, on ClimSim and on the correction-coefficient corpus at nearly half a million rows, the network leads instead, 25.10% against 27.20% and 3.75% against 62.93%, which is the OCO-2 regime. On advection, where the operator is linear, the linear ridge is the lowest row and every other family lands within 0.4 points of it.

Learned metrics help most on the correction-coefficient corpus. From the state alone the isotropic kernel is at 62.93 ± 0.51 and a kernel-flow metric [7] takes it to 3.97 ± 0.42, a factor of sixteen, which brings a kernel on the state within six percent of the network on a corpus where the plain kernel was seventeen times worse. With the low-fidelity coefficients as inputs the same pair reads 15.76 and 4.60, a factor of 3.4: the easier the problem is made for the plain kernel, the less the learned metric adds.

The convex stack is the lowest row on eight of the ten configurations and the second lowest on the other two; the residual-corrected network is among the lowest two on five. Both exceptions are the linear problems, and on both the row that beats the stack is the ridge, which the stack does not contain. On advection the ridge is lower by 0.07 points while the stack, at 11.36%, matches its best member, the isotropic kernel, to two decimals; that kernel and the learned-metric kernel, which differ by less than 0.01 points there, carry between 0.85 and 0.91 of its weight at every split. On the Helmholtz staircase the ridge reaches 0.004%, sixteen times below the next row and about a hundred times below the weakest member of the pool, and the stack reaches 0.066%, slightly below its best member, the learned-metric kernel at 0.067%, on which it puts weight 0.94. In both cases the stack does what it is asked within its pool and cannot recover a family the pool excludes. A combination is a good default when no member of its pool dominates it, and it can be read against the whole comparison only when the pool contains the families being compared.

These rows use one training budget per corpus and one architecture family, and several of the corpora have published results from methods tuned for them. With three to five runs, and one on two rows, the spreads support the large orderings above and not fine distinctions between adjacent families. Every kernel column also inherits the tuning protocol of Section S5, with the length scale chosen on a 6,000-row subsample and the winner refit on every row. On the EMIT table, choosing it with every row in the solve lowers the isotropic kernel's radiance error

by a factor 0.882 on average, so the kernel columns of Table S5 describe the stated training and selection procedures rather than the best each kernel can reach.

A. Configurations

corpus	n	d	q	runs
Advection	19,000	200	200	5
Burgers $\nu = 0.1$	7,200	256	128	3
Burgers $\nu = 0.01$	7,200	256	128	3
Burgers $\nu = 0.001$	7,200	256	128	3
Radiative layer	7,200	256	256	3
Active matter	14,000	256	256	1
Helmholtz staircase	20,384	256	256	1
ClimSim	100,000	124	128	5
RT coefficients, state	487,773–487,823	8	3	5
RT coefficients, +6S	487,773–487,823	11	3	5

B. Reference, input kernels and network

corpus	linear ridge	KRR	learned-metric KRR	network
Advection	11.29±0.04	11.36±0.04	11.36±0.04	11.69±0.07
Burgers $\nu = 0.1$	13.89±0.53	2.81±0.28	2.90±0.33	20.34±0.61
Burgers $\nu = 0.01$	33.68±0.28	23.40±0.54	23.58±0.50	36.65±1.95
Burgers $\nu = 0.001$	38.38±0.12	31.66±0.16	33.23±1.73	39.98±0.65
Radiative layer	31.73	30.31	30.26±0.01	32.17±0.02
Active matter	37.24	35.81	35.18	37.31
Helmholtz staircase	0.00	0.26	0.07	0.40
ClimSim	32.24±0.06	27.20±0.05	26.31±0.04	25.10±0.18
RT coefficients, state	71.54±0.59	62.93±0.51	3.97±0.42	3.75±0.25
RT coefficients, +6S	33.83±0.26	15.76±1.98	4.60±0.31	2.79±0.14

C. Ensembles, corrections and combinations

corpus	ensemble	network + residual KRR	feature kernel	stack
Advection	11.50±0.05	11.50±0.05	11.64±0.07	11.36±0.04
Burgers $\nu = 0.1$	19.28±0.09	2.78±0.29	4.92±0.30	2.78±0.29
Burgers $\nu = 0.01$	35.31±0.70	23.12±0.56	29.20±0.59	23.01±0.54
Burgers $\nu = 0.001$	38.89±0.11	31.82±0.11	35.59±0.48	31.34±0.14
Radiative layer	31.58±0.02	30.40±0.01	31.08±0.01	30.12
Active matter	36.93	35.77	36.22	35.05
Helmholtz staircase	0.24	0.23	0.45	0.07
ClimSim	24.43±0.17	24.33±0.13	24.71±0.25	24.25±0.17
RT coefficients, state	3.24±0.11	3.13±0.05	3.52±0.07	3.09±0.05
RT coefficients, +6S	2.40±0.11	2.34±0.09	2.80±0.09	2.32±0.08

Table S5: Ten further configurations: dimensions and run counts (A), reference models (B), and combinations (C). Errors are mean relative spectral norms in percent, with the 0.05 denominator floor of (S7.1) for the two RT-coefficient rows. Spreads are sample standard deviations; the radiative-layer repetitions share one split and vary initialization, and two rows are single runs. The network budget is 200 epochs at width 512, except ClimSim (60 at 512) and RT coefficients (120 at 384). The linear ridge is excluded from the stack; its Helmholtz error is 0.004%, rounded in the table. Kernel solves above 20,000 training rows use 6,000 Nyström landmarks.

S8 Extended diagnostic tables

This section collects the perturbation, ridge-path, synthetic and external-benchmark tables referred to in Sections S2–S4, and the record of the runs behind Section 6.3 of the article.

width	seed	machine	raw	admissible	matched	validation	test
3×512	101	DGX	18,884	15,287	15,287	2,098	2,331
	102	DGX	18,884	15,239	15,239	2,098	2,331
	103	DGX	18,884	15,284	15,284	2,098	2,331
	104	DGX	18,884	15,286	15,286	2,098	2,331
	105	DGX	18,884	15,271	15,271	2,098	2,331
	106	Kaggle	18,884	15,290	15,290	2,098	2,331
	107	Kaggle	18,884	15,325	15,325	2,098	2,331
	108	Kaggle	18,884	15,250	15,250	2,098	2,331
	109	Kaggle	18,884	15,253	15,253	2,098	2,331
	110	Kaggle	18,884	15,264	15,264	2,098	2,331
3×2000	101	DGX	18,884	15,287	15,287	2,098	2,331
	102	DGX	18,884	15,239	15,239	2,098	2,331
	103	DGX	18,884	15,284	15,284	2,098	2,331
	104	DGX	18,884	15,286	15,286	2,098	2,331
	105	DGX	18,884	15,271	15,271	2,098	2,331
	106	Kaggle	18,884	15,290	15,290	2,098	2,331
	107	Kaggle	18,884	15,325	15,325	2,098	2,331
	108	Kaggle	18,884	15,250	15,250	2,098	2,331
	109	Kaggle	18,884	15,253	15,253	2,098	2,331
	110	Kaggle	18,884	15,264	15,264	2,098	2,331

Table S6: The runs of the training-target experiment: for each split and width, the machine it ran on and the number of training states in each arm. Validation and test blocks are the same in the three arms. The admissible and matched arms have the same size by construction.

Table S7: Same-data perturbation diagnostics for Proposition S1.3. Movement is a Euclidean prediction change in retained coordinates. The four ratio columns are *medians of entrywise bound/movement ratios*, not ratios of medians. The fourteen state-input rows use the direct flux Y_2 at seed 101 with 2,000 fitting rows; the feature rows use four component records at seed 101 and four at seed 104, each with 4,000 fitting and 1,000 query rows. The signed minimum eigenvalue is rounded: a displayed −1.000 does not assert exact equality to −1. A dash means the corresponding bound is not applicable.

source	perturbation	$\lambda_{\min}(B)$	$\|B\|$	median movement	residual-action ratio	signed-spectrum ratio	$\|B\| < 1$ ratio	ridge ratio
state	Nyström 50	-0.941	0.941	2.01	57.3	4.05e+03	4.05e+03	–
state	Nyström 100	-0.872	0.872	1.05	82.1	3.05e+03	3.05e+03	–
state	Nyström 200	-0.642	0.642	0.498	136	1.65e+03	1.65e+03	–
state	Nyström 400	-0.411	0.411	0.2	178	1.54e+03	1.54e+03	–
state	Nyström 800	-0.193	0.193	0.0702	264	1.47e+03	1.47e+03	–
state	Nyström 1600	-0.078	0.078	0.0103	408	3.51e+03	3.51e+03	–
state	ridge ×0.5	-0.500	0.500	0.304	517	1.34e+03	1.34e+03	–
state	ridge ×2.0	+0.000	0.999	0.372	835	550	1.32e+06	550
state	ridge ×10.0	+0.002	8.993	1.36	1.98e+03	271	–	271
state	ridge ×100.0	+0.018	98.918	3.67	7.38e+03	111	–	111
state	PCA inputs 2	-0.998	3.600	10.5	14.3	4.81e+04	–	–
state	PCA inputs 3	-0.998	2.523	9.8	14.2	4.49e+04	–	–
state	PCA inputs 4	-0.998	1.581	5.24	25.3	6.52e+04	–	–
state	PCA inputs 5	-0.997	0.997	1.9	64.3	1.39e+05	1.39e+05	–
s101, Y_1	PCA features 1000	-0.192	0.192	0.00383	4.01e+03	3.8e+04	3.8e+04	–
s101, Y_1	PCA features 500	-0.503	0.503	0.0107	4.41e+03	5.82e+04	5.82e+04	–
s101, Y_1	PCA features 250	-0.764	0.764	0.0185	4.3e+03	1.1e+05	1.1e+05	–
s101, Y_1	PCA features 125	-0.917	0.917	0.0279	3.88e+03	2.6e+05	2.6e+05	–
s101, Y_1	PCA features 64	-0.965	0.965	0.0407	3.55e+03	4.78e+05	4.78e+05	–
s101, Y_1	PCA features 32	-0.991	0.991	0.0595	3.13e+03	1.59e+06	1.59e+06	–
s101, Y_1	PCA features 16	-0.999	0.999	0.0923	2.48e+03	1.3e+07	1.3e+07	–
s101, Y_1	Nyström 100	-1.000	1.000	0.465	637	8.85e+07	8.85e+07	–
s101, Y_1	Nyström 400	-1.000	1.000	0.223	1.27e+03	7.11e+07	7.11e+07	–
s101, Y_1	Nyström 1600	-1.000	1.000	0.0746	2.58e+03	2.74e+07	2.74e+07	–
s101, Y_1	Nyström 3200	-0.997	0.997	0.0196	5.22e+03	9.75e+06	9.75e+06	–
s101, Y_1	ridge ×0.5	-0.490	0.490	0.0102	9.61e+03	5.74e+04	5.74e+04	–
s101, Y_1	ridge ×2.0	+0.000	0.979	0.0115	1.7e+04	2.62e+04	2.5e+06	2.62e+04
s101, Y_1	ridge ×10.0	+0.000	8.813	0.0407	4.32e+04	1.34e+04	–	1.34e+04
s101, Y_1	ridge ×100.0	+0.000	96.942	0.0889	2.17e+05	6.78e+03	–	6.78e+03
s101, Y_2	PCA features 1000	-0.122	0.122	0.00143	6.41e+03	4.95e+04	4.95e+04	–
s101, Y_2	PCA features 500	-0.418	0.418	0.00425	6.8e+03	8.59e+04	8.59e+04	–

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Table S7 (continued)

source	perturbation	$\lambda_{\min}(B)$	$\|B\|$	median movement	residual-action ratio	signed-spectrum ratio	$\|B\| < 1$ ratio	ridge ratio
s101, Y_2	PCA features 250	-0.726	0.726	0.00802	6.45e+03	1.7e+05	1.7e+05	–
s101, Y_2	PCA features 125	-0.868	0.868	0.0117	6.33e+03	2.99e+05	2.99e+05	–
s101, Y_2	PCA features 64	-0.977	0.977	0.0184	5.76e+03	1.3e+06	1.3e+06	–
s101, Y_2	PCA features 32	-0.989	0.989	0.0267	5.78e+03	2.3e+06	2.3e+06	–
s101, Y_2	PCA features 16	-0.999	0.999	0.0419	4.83e+03	1.49e+07	1.49e+07	–
s101, Y_2	Nyström 100	-1.000	1.000	0.345	731	1.01e+08	1.01e+08	–
s101, Y_2	Nyström 400	-1.000	1.000	0.122	1.97e+03	8.48e+07	8.48e+07	–
s101, Y_2	Nyström 1600	-1.000	1.000	0.0823	2.38e+03	5.34e+07	5.34e+07	–
s101, Y_2	Nyström 3200	-0.998	0.998	0.0161	7.05e+03	1.86e+07	1.86e+07	–
s101, Y_2	ridge ×0.5	-0.491	0.491	0.00632	1.71e+04	7.71e+04	7.71e+04	–
s101, Y_2	ridge ×2.0	+0.000	0.982	0.00687	3.14e+04	3.64e+04	3.96e+06	3.64e+04
s101, Y_2	ridge ×10.0	+0.000	8.836	0.0249	7.8e+04	1.82e+04	–	1.82e+04
s101, Y_2	ridge ×100.0	+0.000	97.199	0.0545	3.92e+05	9.17e+03	–	9.17e+03
s101, Y_3	PCA features 1000	-0.170	0.170	0.00292	6.17e+03	4.56e+04	4.56e+04	–
s101, Y_3	PCA features 500	-0.475	0.475	0.00766	7.02e+03	7.73e+04	7.73e+04	–
s101, Y_3	PCA features 250	-0.738	0.738	0.0136	6.44e+03	1.38e+05	1.38e+05	–
s101, Y_3	PCA features 125	-0.898	0.898	0.0211	5.61e+03	2.87e+05	2.87e+05	–
s101, Y_3	PCA features 64	-0.964	0.964	0.028	5.45e+03	7.06e+05	7.06e+05	–
s101, Y_3	PCA features 32	-0.990	0.990	0.0406	4.9e+03	2.09e+06	2.09e+06	–
s101, Y_3	PCA features 16	-0.999	0.999	0.0725	3.51e+03	1.63e+07	1.63e+07	–
s101, Y_3	Nyström 100	-1.000	1.000	0.35	876	4.97e+07	4.97e+07	–
s101, Y_3	Nyström 400	-1.000	1.000	0.191	1.55e+03	4.53e+07	4.53e+07	–
s101, Y_3	Nyström 1600	-0.999	0.999	0.0709	3.33e+03	1.45e+07	1.45e+07	–
s101, Y_3	Nyström 3200	-0.997	0.997	0.0331	4.54e+03	6.39e+06	6.39e+06	–
s101, Y_3	ridge ×0.5	-0.492	0.492	0.00865	1.65e+04	7.26e+04	7.26e+04	–
s101, Y_3	ridge ×2.0	+0.000	0.985	0.00949	3e+04	3.38e+04	4.43e+06	3.38e+04
s101, Y_3	ridge ×10.0	+0.000	8.864	0.0357	7.18e+04	1.63e+04	–	1.63e+04
s101, Y_3	ridge ×100.0	+0.000	97.500	0.0821	3.44e+05	7.81e+03	–	7.81e+03
s101, Y_4	PCA features 1000	-0.006	0.006	0.00138	538	4.64e+03	4.64e+03	–
s101, Y_4	PCA features 500	-0.032	0.032	0.00463	712	7.83e+03	7.83e+03	–
s101, Y_4	PCA features 250	-0.114	0.114	0.0104	826	1.35e+04	1.35e+04	–
s101, Y_4	PCA features 125	-0.269	0.269	0.0191	897	2.12e+04	2.12e+04	–
s101, Y_4	PCA features 64	-0.594	0.594	0.0286	963	5.8e+04	5.8e+04	–
s101, Y_4	PCA features 32	-0.741	0.741	0.0436	983	8.06e+04	8.06e+04	–
s101, Y_4	PCA features 16	-0.902	0.902	0.0769	987	1.65e+05	1.65e+05	–
s101, Y_4	Nyström 100	-0.998	0.998	0.603	304	1.2e+06	1.2e+06	–

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Table S7 (continued)

source	perturbation	$\lambda_{\min}(B)$	$\|B\|$	median movement	residual-action ratio	signed-spectrum ratio	$\|B\| < 1$ ratio	ridge ratio
s101, Y_4	Nyström 400	-0.985	0.985	0.35	500	2.11e+05	2.11e+05	–
s101, Y_4	Nyström 1600	-0.860	0.860	0.135	916	4.86e+04	4.86e+04	–
s101, Y_4	Nyström 3200	-0.762	0.762	0.0397	1.71e+03	8.46e+04	8.46e+04	–
s101, Y_4	ridge ×0.5	-0.494	0.494	0.063	3.15e+03	1.61e+04	1.61e+04	–
s101, Y_4	ridge ×2.0	+0.000	0.989	0.0599	6.6e+03	8.63e+03	1.56e+06	8.63e+03
s101, Y_4	ridge ×10.0	+0.000	8.901	0.191	1.86e+04	4.89e+03	–	4.89e+03
s101, Y_4	ridge ×100.0	+0.000	97.908	0.36	1.07e+05	2.86e+03	–	2.86e+03
s104, Y_1	PCA features 1000	-0.184	0.184	0.00397	4.68e+03	3.85e+04	3.85e+04	–
s104, Y_1	PCA features 500	-0.471	0.471	0.0104	5.01e+03	5.82e+04	5.82e+04	–
s104, Y_1	PCA features 250	-0.724	0.724	0.0175	4.8e+03	1.04e+05	1.04e+05	–
s104, Y_1	PCA features 125	-0.901	0.901	0.0267	4.25e+03	2.45e+05	2.45e+05	–
s104, Y_1	PCA features 64	-0.959	0.959	0.0373	3.87e+03	4.76e+05	4.76e+05	–
s104, Y_1	PCA features 32	-0.985	0.985	0.055	3.43e+03	1.07e+06	1.07e+06	–
s104, Y_1	PCA features 16	-0.997	0.997	0.0893	2.61e+03	3.51e+06	3.51e+06	–
s104, Y_1	concatenation	-0.749	76.009	0.0427	2.3e+05	1.99e+05	–	–
s104, Y_1	member 1	-0.973	175.948	0.049	2.34e+05	2.24e+06	–	–
s104, Y_1	member 2	-0.967	126.717	0.0541	3.81e+05	2.02e+06	–	–
s104, Y_1	Nyström 100	-1.000	1.000	0.861	379	9.61e+07	9.61e+07	–
s104, Y_1	Nyström 400	-1.000	1.000	0.247	1.25e+03	6.69e+07	6.69e+07	–
s104, Y_1	Nyström 1600	-1.000	1.000	0.112	2.01e+03	4.7e+07	4.7e+07	–
s104, Y_1	Nyström 3200	-1.000	1.000	0.0249	4.43e+03	5.86e+07	5.86e+07	–
s104, Y_1	ridge ×0.5	-0.493	0.493	0.011	1.06e+04	5.96e+04	5.96e+04	–
s104, Y_1	ridge ×2.0	+0.000	0.986	0.0124	1.89e+04	2.7e+04	3.93e+06	2.7e+04
s104, Y_1	ridge ×10.0	+0.000	8.877	0.0445	4.73e+04	1.36e+04	–	1.36e+04
s104, Y_1	ridge ×100.0	+0.000	97.646	0.102	2.27e+05	6.54e+03	–	6.54e+03
s104, Y_2	PCA features 1000	-0.108	0.108	0.00163	7.72e+03	5.02e+04	5.02e+04	–
s104, Y_2	PCA features 500	-0.370	0.370	0.00553	8.84e+03	7.23e+04	7.23e+04	–
s104, Y_2	PCA features 250	-0.668	0.668	0.0107	8.77e+03	1.29e+05	1.29e+05	–
s104, Y_2	PCA features 125	-0.834	0.834	0.0166	7.71e+03	2.12e+05	2.12e+05	–
s104, Y_2	PCA features 64	-0.955	0.955	0.0234	6.59e+03	6.62e+05	6.62e+05	–
s104, Y_2	PCA features 32	-0.983	0.983	0.0326	6.12e+03	1.46e+06	1.46e+06	–
s104, Y_2	PCA features 16	-0.998	0.998	0.0503	5.04e+03	1.05e+07	1.05e+07	–
s104, Y_2	concatenation	-0.734	96.255	0.0279	6.7e+05	3.68e+05	–	–
s104, Y_2	member 1	-0.974	200.331	0.0328	8.4e+05	5.59e+06	–	–
s104, Y_2	member 2	-0.968	266.674	0.035	9.16e+05	3.4e+06	–	–
s104, Y_2	Nyström 100	-1.000	1.000	0.713	451	1.85e+08	1.85e+08	–

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Table S7 (continued)

source	perturbation	$\lambda_{\min}(B)$	$\|B\|$	median movement	residual-action ratio	signed-spectrum ratio	$\|B\| < 1$ ratio	ridge ratio
s104, Y_2	Nyström 400	-1.000	1.000	0.119	2.5e+03	4.07e+07	4.07e+07	–
s104, Y_2	Nyström 1600	-0.999	0.999	0.0322	6.59e+03	1.8e+07	1.8e+07	–
s104, Y_2	Nyström 3200	-0.992	0.992	0.0104	1.25e+04	8.1e+06	8.1e+06	–
s104, Y_2	ridge ×0.5	-0.498	0.498	0.00788	2.13e+04	8.49e+04	8.49e+04	–
s104, Y_2	ridge ×2.0	+0.000	0.995	0.00862	3.89e+04	3.91e+04	1.66e+07	3.91e+04
s104, Y_2	ridge ×10.0	+0.000	8.958	0.0323	9.36e+04	1.88e+04	–	1.88e+04
s104, Y_2	ridge ×100.0	+0.000	98.534	0.0733	4.53e+05	9.13e+03	–	9.13e+03
s104, Y_3	PCA features 1000	-0.176	0.176	0.00305	6.26e+03	4.48e+04	4.48e+04	–
s104, Y_3	PCA features 500	-0.474	0.474	0.00801	7.06e+03	7.26e+04	7.26e+04	–
s104, Y_3	PCA features 250	-0.738	0.738	0.0145	6.18e+03	1.26e+05	1.26e+05	–
s104, Y_3	PCA features 125	-0.882	0.882	0.0189	6.1e+03	2.67e+05	2.67e+05	–
s104, Y_3	PCA features 64	-0.957	0.957	0.0259	5.75e+03	6.3e+05	6.3e+05	–
s104, Y_3	PCA features 32	-0.990	0.990	0.0371	5.12e+03	2.16e+06	2.16e+06	–
s104, Y_3	PCA features 16	-0.999	0.999	0.0648	3.65e+03	2.51e+07	2.51e+07	–
s104, Y_3	concatenation	-0.749	12.229	0.0228	6.41e+04	2.52e+05	–	–
s104, Y_3	member 1	-0.987	28.650	0.0374	1.01e+05	5.18e+06	–	–
s104, Y_3	member 2	-0.978	32.067	0.0391	3.02e+04	2.59e+06	–	–
s104, Y_3	Nyström 100	-1.000	1.000	0.777	389	1.27e+08	1.27e+08	–
s104, Y_3	Nyström 400	-1.000	1.000	0.203	1.42e+03	5.4e+07	5.4e+07	–
s104, Y_3	Nyström 1600	-1.000	1.000	0.134	1.72e+03	4.38e+07	4.38e+07	–
s104, Y_3	Nyström 3200	-0.999	0.999	0.029	4.4e+03	1.58e+07	1.58e+07	–
s104, Y_3	ridge ×0.5	-0.487	0.487	0.00833	1.62e+04	7.25e+04	7.25e+04	–
s104, Y_3	ridge ×2.0	+0.000	0.975	0.00917	2.93e+04	3.42e+04	2.65e+06	3.42e+04
s104, Y_3	ridge ×10.0	+0.000	8.771	0.0329	7.36e+04	1.73e+04	–	1.73e+04
s104, Y_3	ridge ×100.0	+0.000	96.479	0.0765	3.48e+05	8.23e+03	–	8.23e+03
s104, Y_4	PCA features 1000	-0.003	0.003	0.000765	496	4.96e+03	4.96e+03	–
s104, Y_4	PCA features 500	-0.019	0.019	0.00303	682	8.81e+03	8.81e+03	–
s104, Y_4	PCA features 250	-0.088	0.088	0.00823	778	1.58e+04	1.58e+04	–
s104, Y_4	PCA features 125	-0.259	0.259	0.0153	851	3.11e+04	3.11e+04	–
s104, Y_4	PCA features 64	-0.406	0.406	0.0238	906	4e+04	4e+04	–
s104, Y_4	PCA features 32	-0.790	0.790	0.0427	859	1.27e+05	1.27e+05	–
s104, Y_4	PCA features 16	-0.883	0.883	0.0733	851	1.62e+05	1.62e+05	–
s104, Y_4	concatenation	-0.658	58.594	0.0605	1.49e+04	1.97e+05	–	–
s104, Y_4	member 1	-0.897	23.688	0.103	5.19e+03	3.18e+05	–	–
s104, Y_4	member 2	-0.919	155.529	0.108	1.74e+04	8.07e+05	–	–
s104, Y_4	Nyström 100	-0.993	0.993	0.517	378	3.93e+05	3.93e+05	–

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Table S7 (continued)

source	perturbation	$\lambda_{\min}(B)$	$\|B\|$	median movement	residual-action ratio	signed-spectrum ratio	$\|B\| < 1$ ratio	ridge ratio
s104, Y_4	Nyström 400	-0.961	0.961	0.241	772	1.39e+05	1.39e+05	–
s104, Y_4	Nyström 1600	-0.850	0.850	0.0775	1.66e+03	9.85e+04	9.85e+04	–
s104, Y_4	Nyström 3200	-0.707	0.707	0.0213	2.57e+03	1.5e+05	1.5e+05	–
s104, Y_4	ridge ×0.5	-0.491	0.491	0.0482	4.89e+03	2.63e+04	2.63e+04	–
s104, Y_4	ridge ×2.0	+0.000	0.983	0.0525	8.97e+03	1.24e+04	1.43e+06	1.24e+04
s104, Y_4	ridge ×10.0	+0.000	8.846	0.179	2.36e+04	6.6e+03	–	6.6e+03
s104, Y_4	ridge ×100.0	+0.000	97.301	0.393	1.18e+05	3.32e+03	–	3.32e+03

Table S8: Ridge-path diagnostics, seed 101, 4,000 fitting and 1,000 query rows. The bias and noise factor refer to Proposition S1.1(iii); the factor multiplies a hypothetical noise variance and is not an estimate of simulator noise. Corrected and kernel-only test columns are mean relative errors of a physical spectral component, in percent. The final column uses retained PCA coordinates on the held-out half and compares same-half fitting with split-half fitting; it is a different metric and sample. No ridge is selected on these test values.

component	kernel	γ	$\mathrm{tr}S_\lambda$	bias	noise factor	median $\|a_\lambda\|^2$	median $\widetilde{P}_\lambda$	corrected [%]	kernel only [%]	same / split half PCA error [%]
Y1	input	1e-08	3841	0.000226	0.924	0.87	0.0324	0.2168	0.1754	0.079 / 2.465
Y1	input	1e-06	1624	0.0712	0.26	0.226	0.0395	0.2376	0.2420	1.145 / 2.214
Y1	input	0.0001	212	0.282	0.0269	0.0237	0.0997	0.4132	0.9768	2.221 / 2.395
Y1	input	0.01	19	0.413	0.00194	0.00197	0.315	0.5095	5.7578	2.716 / 2.728
Y1	feature	1e-08	2541	0.0012	0.51	0.481	0.00637	0.2027	0.2111	0.280 / 1.171
Y1	feature	1e-06	689	0.0225	0.111	0.0668	0.0151	0.3130	0.3275	1.085 / 1.464
Y1	feature	0.0001	103	0.156	0.0137	0.00825	0.0529	0.4546	0.6396	2.105 / 2.187
Y1	feature	0.01	9	0.418	0.001	0.000845	0.185	0.5326	5.9929	2.783 / 2.786
Y2	input	1e-08	3884	0.000107	0.943	0.523	0.0369	0.4559	0.6885	0.043 / 1.784
Y2	input	1e-06	1554	0.061	0.219	0.158	0.0416	0.4849	0.9183	0.886 / 1.632
Y2	input	0.0001	132	0.211	0.0143	0.0127	0.0845	0.6325	4.4719	1.516 / 1.580
Y2	input	0.01	10	0.273	0.00115	0.00117	0.227	0.8581	26.9965	1.773 / 1.777
Y2	feature	1e-08	2339	0.000906	0.451	0.44	0.00562	0.1431	0.1439	0.184 / 0.738
Y2	feature	1e-06	564	0.0159	0.0893	0.0618	0.0143	0.2768	0.2696	0.662 / 0.899
Y2	feature	0.0001	86	0.134	0.0118	0.00829	0.0512	0.5764	1.0187	1.275 / 1.326
Y2	feature	0.01	9	0.27	0.000998	0.000909	0.188	0.8905	15.5377	1.738 / 1.741
Y3	input	1e-08	3308	0.00797	0.7	0.412	0.0132	0.2382	0.3731	0.324 / 2.147
Y3	input	1e-06	558	0.283	0.0644	0.0523	0.0192	0.2627	0.9135	1.656 / 1.951
Y3	input	0.0001	45	0.455	0.00546	0.00501	0.0457	0.3432	5.4172	2.173 / 2.155
Y3	input	0.01	5	0.554	0.000642	0.000643	0.152	0.4914	19.3748	2.715 / 2.702
Y3	feature	1e-08	2371	0.00158	0.461	0.458	0.00569	0.0999	0.1197	0.221 / 0.903
Y3	feature	1e-06	590	0.0255	0.094	0.063	0.0145	0.1650	0.1866	0.829 / 1.136
Y3	feature	0.0001	90	0.267	0.0123	0.00842	0.0513	0.3145	0.6620	1.790 / 1.882
Y3	feature	0.01	9	0.525	0.001	0.000885	0.189	0.4913	7.4936	2.612 / 2.624
Y4	input	1e-08	3998	9.17e-07	0.999	0.473	0.269	2.1826	2.3706	0.007 / 12.844
Y4	input	1e-06	3792	0.00736	0.9	0.437	0.27	2.1557	2.3495	0.629 / 12.779
Y4	input	0.0001	1074	1.16	0.125	0.0912	0.319	1.9633	2.4907	7.986 / 11.573
Y4	input	0.01	51	2.3	0.00312	0.00306	0.619	2.3401	8.9051	12.340 / 12.554
Y4	feature	1e-08	3823	0.00807	0.917	0.975	0.031	2.4285	2.4310	0.626 / 14.459
Y4	feature	1e-06	1652	0.794	0.272	0.24	0.0392	1.8004	1.8078	6.037 / 11.377
Y4	feature	0.0001	229	1.83	0.0293	0.0257	0.103	1.8454	1.9424	9.636 / 10.497
Y4	feature	0.01	19	2.39	0.0019	0.00181	0.323	2.4706	4.7750	12.907 / 12.992

β	average-family measured slope	$\beta/(\beta+2)$	uniform-family measured slope	analytic slope same ladder
0.5	0.200	0.200	0.494	0.494
1	0.333	0.333	0.973	0.973
2	0.498	0.500	1.721	1.724
3	0.597	0.600	1.963	1.963
4	0.664	0.667	1.995	1.995

β	uniform-family maximum relative gap	profile-family maximum relative gap	violations
0.5	5.7e-04	4.4e-04	0
1	1.7e-03	3.9e-04	0
2	1.0e-02	3.1e-04	0
3	1.4e-03	5.2e-04	0
4	2.8e-04	2.8e-04	0

Table S9: Manufactured transfer families, seed zero, two million samples for each β. The first panel compares finite-ladder slopes with the analytic curve on the same ladder. The second reports the largest relative discrepancies between the Monte Carlo mean squared error and the analytic value (uniform family) or the quadrature value (profile family), and the total number of pointwise violations.

Table S10: Matched correction-coefficient benchmark [2]. The standard split uses 35,000 / 7,500 / 7,500 states. OOD test states exceed the 0.85 aerosol or water-vapor quantile; remaining states are split 85/15. Cells average three seeds or initializations; the reference rows are the published values. RMSE, MAE and SMAPE are pooled; R^2 is the uniform average over the three coefficients. Model keys: dkr, feature kernel; krr, input kernel; mlp, network; ens, ensemble; resid, residual correction; lowfi, the low-fidelity mean. The categorical inputs follow the benchmark protocol.

split	inputs	head	RMSE	MAE	R^2	SMAPE [%]
OOD	state only	dkr	0.01408±0.00177	0.00709±0.00083	0.99226±0.00106	5.714±0.441
OOD	state only	krr	0.11283±0.00069	0.05430±0.00197	0.66968±0.01490	30.762±1.160
OOD	state only	mlp	0.00750±0.00044	0.00355±0.00036	0.99586±0.00012	4.500±0.249
OOD	state only	mlp_ens	0.00624±0.00015	0.00258±0.00017	0.99667±0.00004	3.470±0.370
OOD	state only	mlp_resid	0.00620±0.00015	0.00252±0.00015	0.99670±0.00005	3.372±0.308
OOD	state only	ridge	0.11944±0.00000	0.06195±0.00000	0.49009±0.00000	37.228±0.000
OOD	with 6S inputs	dkr	0.00735±0.00036	0.00336±0.00033	0.99597±0.00022	4.101±0.474
OOD	with 6S inputs	krr	0.02904±0.00015	0.01432±0.00005	0.97160±0.00020	11.238±0.033
OOD	with 6S inputs	lowfi_resid	0.02901±0.00016	0.01430±0.00006	0.97172±0.00021	11.416±0.047
OOD	with 6S inputs	mlp	0.00741±0.00080	0.00347±0.00063	0.99602±0.00048	4.406±0.366
OOD	with 6S inputs	mlp_ens	0.00611±0.00030	0.00246±0.00023	0.99682±0.00013	3.009±0.093
OOD	with 6S inputs	mlp_resid	0.00596±0.00031	0.00232±0.00022	0.99687±0.00012	2.921±0.067
OOD	with 6S inputs	ridge	0.08048±0.00000	0.04396±0.00000	0.79274±0.00000	29.064±0.000
standard	state only	dkr	0.00874±0.00121	0.00387±0.00071	0.99379±0.00053	5.342±0.323
standard	state only	krr	0.11288±0.00001	0.05157±0.00002	0.68464±0.00028	31.178±0.032
standard	state only	mlp	0.00657±0.00026	0.00242±0.00030	0.99465±0.00007	5.136±0.550
standard	state only	mlp_ens	0.00591±0.00007	0.00170±0.00005	0.99513±0.00010	3.873±0.132
standard	state only	mlp_resid	0.00581±0.00004	0.00151±0.00004	0.99520±0.00010	3.543±0.066
standard	state only	ridge	0.11959±0.00000	0.06067±0.00000	0.48572±0.00000	41.647±0.000
standard	with 6S inputs	dkr	0.00627±0.00025	0.00218±0.00021	0.99502±0.00028	4.429±0.169
standard	with 6S inputs	krr	0.02514±0.00010	0.01155±0.00003	0.97447±0.00010	10.184±0.037
standard	with 6S inputs	lowfi_resid	0.02513±0.00010	0.01155±0.00003	0.97452±0.00010	10.178±0.041
standard	with 6S inputs	mlp	0.00632±0.00018	0.00224±0.00017	0.99504±0.00014	4.496±0.160
standard	with 6S inputs	mlp_ens	0.00587±0.00014	0.00163±0.00012	0.99531±0.00011	3.407±0.130
standard	with 6S inputs	mlp_resid	0.00578±0.00012	0.00148±0.00008	0.99535±0.00010	3.185±0.097
standard	with 6S inputs	ridge	0.06981±0.00000	0.03631±0.00000	0.83813±0.00000	28.300±0.000
standard	published	pKANrtm	0.00619	0.00186	0.99472	3.912
OOD	published	pKANrtm	0.01022	0.00544	0.99418	5.399

Table S11: OCO-2 selection comparison, seeds 0–2 on each of three bands. Each record uses 2,000 validation and 2,000 test rows. Both reported columns are *mean relative norms* in percent; the stack fitting objectives are row-relative *squared* errors. The radiance quadratic uses the exact affine reconstruction Gram matrix. Model keys: dkr, feature kernel; mean, trained network; ridge, linear readout; raw/ard, input kernels. Flat, radx and wnum are the three training losses. Selection is per coordinate, with the radiance variant using greedy coordinate updates, not exhaustive global search.

band	head / rule	reduced error [%]	radiance error [%]
o2	select_radiance	4.08±0.05	0.0283±0.0017
o2	select_reduced	4.09±0.06	0.0283±0.0017
o2	single dkr_flat	4.07±0.06	0.0757±0.0012
o2	single dkr_radx	20.14±0.07	0.0351±0.0006
o2	single dkr_wnum	21.70±0.60	0.0309±0.0017
o2	single kernel_ard	30.02±0.03	0.2706±0.1030
o2	single kernel_raw	40.53±0.09	0.2353±0.0047
o2	single mean_flat	4.40±0.08	0.1756±0.0111
o2	single mean_radx	58.72±0.60	0.0510±0.0049
o2	single mean_wnum	49.22±0.34	0.0437±0.0002
o2	single ridge_flat	4.67±0.09	0.2205±0.0124
o2	single ridge_radx	44.83±0.43	0.0538±0.0044
o2	single ridge_wnum	45.55±0.73	0.0527±0.0016
o2	stack_radiance	18.59±0.25	0.0264±0.0007
o2	stack_reduced	4.10±0.07	0.0291±0.0029
sco2	select_radiance	8.09±0.02	0.0598±0.0013
sco2	select_reduced	8.09±0.02	0.0598±0.0013
sco2	single dkr_flat	8.09±0.02	0.1083±0.0114
sco2	single dkr_radx	15.27±0.25	0.0780±0.0094
sco2	single dkr_wnum	15.48±0.31	0.0638±0.0022
sco2	single kernel_ard	28.72±0.02	0.6023±0.0097
sco2	single kernel_raw	41.01±0.09	0.5229±0.0081
sco2	single mean_flat	8.23±0.01	0.2107±0.0194
sco2	single mean_radx	46.08±0.91	0.0936±0.0082
sco2	single mean_wnum	36.64±0.38	0.0750±0.0020
sco2	single ridge_flat	8.34±0.02	0.2568±0.0244
sco2	single ridge_radx	41.14±1.78	0.0967±0.0081
sco2	single ridge_wnum	40.25±0.24	0.0887±0.0034
sco2	stack_radiance	8.87±0.15	0.0511±0.0016
sco2	stack_reduced	8.09±0.01	0.0536±0.0020
wco2	select_radiance	16.26±0.07	0.0510±0.0086
wco2	select_reduced	16.26±0.07	0.0510±0.0086
wco2	single dkr_flat	16.26±0.07	0.1135±0.0069
wco2	single dkr_radx	48.28±0.28	0.0475±0.0055
wco2	single dkr_wnum	48.60±0.38	0.0516±0.0086
wco2	single kernel_ard	52.45±0.07	0.2231±0.0014
wco2	single kernel_raw	58.63±0.20	0.4431±0.0063
wco2	single mean_flat	16.42±0.06	0.2254±0.0161
wco2	single mean_radx	67.92±0.46	0.0605±0.0075
wco2	single mean_wnum	62.39±0.42	0.0700±0.0087
wco2	single ridge_flat	16.49±0.07	0.2571±0.0234
wco2	single ridge_radx	61.03±0.31	0.0663±0.0080
wco2	single ridge_wnum	61.33±0.05	0.0778±0.0095
wco2	stack_radiance	37.43±0.20	0.0391±0.0026
wco2	stack_reduced	16.28±0.07	0.0404±0.0034

S9 Proofs

Numbers without the prefix S refer to the article.

Proof of Proposition 1. Subtract the two forward maps and use $1/\widehat{q} - 1/q = \rho e_s/(q\widehat{q})$ to obtain (4). Since $u = \rho D$, $\widehat{D} = D + e_t + u e_s - \widehat{s} e_a$ and $\widehat{\rho} - \rho = (u - e_a - \rho\widehat{D})/\widehat{D}$, which gives (5). The triangle inequality yields $|\widehat{D} - D| \le B$ and then (6). □

Proof of Proposition 2. Subtract $h + S(f - h)$ from f to obtain (9). On the design, $I - K(K + \lambda I)^{-1} = \lambda(K + \lambda I)^{-1}$, which proves (10) and its spectral form. Apply the assumed uniform inequality to $g = f - h$ for (11). □

Proof of Proposition 3. For a common operator the corrected error is $a - b$, so expansion of its squared norm gives (12). The error of p_t is $a - tb$; minimize the resulting quadratic on $[0, 1]$ to obtain (13). For distinct operators, $f - h - S_1(f - h) = (I - S_0)f - [(S_1 - S_0)f + (I - S_1)h]$, after which the same expansion applies. □

Proof of Proposition S1.1. (i) is the chain rule applied to $\widehat{y} = S_\lambda y + (I - S_\lambda)h_X(y)$, and the two examples are direct substitutions. (ii) With the mean and kernel frozen the map is affine with linear part S_λ; the trace is the sum of the eigenvalues $\mu_j/(\mu_j + \lambda)$, and the operator norm of a symmetric positive semidefinite matrix is its largest eigenvalue. (iii) $\widehat{y} - f_X = -(I - S_\lambda)r + S_\lambda\varepsilon$; the cross term has zero mean, and both matrices are diagonal in the eigenbasis of K. (iv) The correction at x is $a_\lambda(x)^\top(y - h_X)$, linear in ε with coefficient vector $a_\lambda(x)$. □

Proof of Proposition S1.2. Let $\widehat{\mathcal{R}}_j$ and $\mathcal{R}_j$ be the empirical and population risks. Hoeffding's inequality [25] and a union bound give $|\widehat{\mathcal{R}}_j - \mathcal{R}_j| \le B\sqrt{\log(2M/\delta)/(2m)}$ simultaneously for all candidates with probability at least $1 - \delta$. Apply this once to the empirical minimizer and once to a population minimizer, and use empirical optimality between the two inequalities. □

Proof of Theorem 4. Write $q = 1 - \rho s$, $u = L - a = \rho t/q$, $D = t/q$, $\widehat{D} = \widehat{t} + \widehat{s}(L - \widehat{a})$ and $z = \bar{\rho}$. Suppose first $\widehat{D} > 0$. The map $F(v) = \widehat{D}v - (L - \widehat{a})$ is increasing and affine, z is the projection of its root onto $[0, R]$ and $\rho \in [0, R]$, so $(z - \rho)F(z) \le 0$: at an interior root $F(z) = 0$; at $z = R$ the root exceeds R, so $F(R) < 0$ and $z - \rho \ge 0$; at $z = 0$ the root is negative, so $F(0) > 0$ and $z - \rho \le 0$. Since $L - \widehat{a} = u - e_a$ and, by (5), $\widehat{D} = D + e_t + u e_s - \widehat{s} e_a$ with $u = \rho D$,

$$F(z) = D(z - \rho) + (1 - z\widehat{s})e_a + z e_t + z u e_s,$$

hence $D(z - \rho)^2 \le |z - \rho|\big((1 - z\widehat{s})|e_a| + z|e_t| + zu|e_s|\big)$ and, with $D = t/q$,

$$|z - \rho| \le q(1 - z\widehat{s})\frac{|e_a|}{t} + qz\frac{|e_t|}{t} + z\rho\,|e_s|.$$

Now $q(1 - z\widehat{s})$ is positive and at most 1 under the hypotheses, which is all that is used here, $qz \le R$ and $z\rho \le R^2$, so $|z - \rho| \le e_R/t$; and $|z - \rho| \le R$ because both lie in $[0, R]$. If $\widehat{D} \le 0$, then $\widehat{t} \le \widehat{s}(e_a - u) \le \widehat{s}|e_a|$ and $t = \widehat{t} - e_t \le |e_t| + \widehat{s}|e_a|$, so $Rt \le R|e_t| + R\widehat{s}|e_a| \le R|e_t| + |e_a| \le e_R$; the fallback value and ρ lie in $[0, R]$, so $|z - \rho| \le R = \min\{R, e_R/t\}$. For optimality take $s = \widehat{s} = 0$, $\widehat{t} = t$, $\rho = r_0$ and $\widehat{a} = a + ct$ with $0 < c < r_0 < R$: then $|\bar{\rho} - \rho| = c = e_R/t$ exactly, with a positive denominator and no projection active. This proves (i). For (ii) integrate (i), split at $t = \tau$ and bound e_R^2/t^2 by e_R^2/τ^2 on $\{t > \tau\}$. For (iii), $\varepsilon_R = 0$ gives zero error by (i). Otherwise take $\tau = \varepsilon_R^{2/(\beta+2)} \le u_0$; then $R^2 A\tau^\beta + \varepsilon_R^2/\tau^2 = (R^2 A + 1)\varepsilon_R^{2\beta/(\beta+2)}$. (iv) is (i) with $t \ge t_0$. □

Proof of Proposition 5. $L = r_0 t$ and $\widehat{\rho} = (r_0 t - \widehat{a})/t$; $e_R = |\widehat{a}| = ct\,\mathbf{1}\{t \le \tau\}$; with density $\beta u^{\beta-1}$ on $(0,1)$, $\mathbb{E}_P[t^2\mathbf{1}\{t \le \tau\}] = \beta\tau^{\beta+2}/(\beta+2)$; the retrieval error equals c exactly on $\{t \le \tau\}$, an event of probability τ^β. Eliminate τ. □

Proof of Theorem 6. If $R = 0$ the statement is immediate; assume $R > 0$. Conditional on t the squared error is at most both R^2 and e_R^2/t^2 by Theorem 4(i), so $\mathbb{E}[|\bar{\rho} - \rho|^2 \mid t] \le \min\{R^2, \mathbb{E}[e_R^2 \mid t]/t^2\} \le \min\{R^2, \sigma^2 t^{2\nu-2}\}$; integrate. With $t_* = (\sigma/R)^{1/(1-\nu)}$, $\mathbb{E}_P \min\{R^2, \sigma^2 t^{2\nu-2}\} = R^2 F_t(t_*) + \sigma^2 \mathbb{E}_P[t^{2\nu-2}\mathbf{1}\{t > t_*\}]$; the first term is at most $R^2 A t_*^\beta = O(\sigma^{\beta/(1-\nu)})$ once $t_* \le u_0$, and $\mathbb{E}_P[t^{2\nu-2}\mathbf{1}\{t > t_*\}] = \int_0^{t_*^{2\nu-2}} \mathbb{P}(t_* < t < v^{-1/(2-2\nu)})\,dv \le u_0^{2\nu-2} + A\int_{u_0^{2\nu-2}}^{t_*^{2\nu-2}} v^{-\beta/(2-2\nu)}\,dv$, which is $O(1)$ for $\beta > 2(1-\nu)$, $O(\log(1/\sigma))$ for $\beta = 2(1-\nu)$ and $O(t_*^{\beta-2(1-\nu)})$ for $\beta < 2(1-\nu)$. At $\nu = 1$ the bound is σ^2 before any tail enters. In the model, $c^{-2}\mathbb{E}_P|\bar{\rho} - \rho|^2 = \mathbb{E}_P \min\{1, (\sigma/c)^2 t^{2\nu-2}\}$, the same integral with equality. □

Proof of Proposition 7. (i) With $a = s = 0$ and exact t and s, the stack's path-radiance error is $e(w) = ((1-w)b,\, wc\delta)$, and the radiance at ρ is predicted with the same error, since $q = \widehat{q} = 1$; this gives F. The constrained retrieval evaluates $(L - \widehat{a})/t = \rho - e_i/t_i$ at entry i, with denominator $t_i > 0$, and $|e_i|/t_i \le \max\{b, c\} \le \min\{\rho, R - \rho\}$ keeps it in $[0, R]$, so the projection is inactive and $\bar{\rho} - \rho = -((1-w)b,\, wc)$; this gives E. Both are convex quadratics in w, and setting their derivatives to zero gives w_f and w_r. Substituting,

$$E(w_f) = \frac{b^2c^2\,(c^2\delta^4 + b^2)}{2(b^2 + c^2\delta^2)^2}, \qquad E(w_r) = \frac{b^2c^2}{2(b^2 + c^2)},$$

whose ratio is the one stated. It tends to $(b^2 + c^2)/b^2$ as $\delta \to 0$; at $c = b/\delta$ the two factors of the numerator are $b^2(1 + \delta^2)$ and $b^2(1 + \delta^2)/\delta^2$ and the denominator is $4b^4$. Since w_f minimizes F, $F(w_f) \le F(w_r)$. Finally, with $\widetilde{e}_t = \widetilde{e}_s = 0$ and $\tau \le \delta \le 1$, the weights $(t_i \vee \tau)^{-2}$ in (18) are 1 and δ^{-2}, so $J_\tau(w) = \frac{1}{2}[(1-w)^2 b^2 + w^2 c^2 \delta^2/\delta^2] = E(w)$.

(ii) With $s = \widehat{s} = 0$ and $\widehat{t} = t$, the predicted radiance at ρ is $\widehat{a} + \rho t$, so its error is the path-radiance error r_w of the combination, and the retrieval evaluates $(L - \widehat{a})/t = \rho - r_w/t$, which lies in $[0, R]$ by hypothesis. Hence $F(w) = \mathbb{E}_P r_w^2$ and $E(w) = \mathbb{E}_P[r_w^2/t^2]$. Since the weights sum to one, r_w lies in $\mathcal{L}$, so $\lambda_{\min} F(w) \le E(w) \le \lambda_{\max} F(w)$ for every w (both sides vanish when $\mathbb{E}_P r_w^2 = 0$), and

$$E(w_f) \le \lambda_{\max} F(w_f) \le \lambda_{\max} F(w_r) \le \frac{\lambda_{\max}}{\lambda_{\min}}\, E(w_r).$$

If $t_{\min} \le t \le t_{\max}$ almost surely, then $r^2/t_{\max}^2 \le r^2/t^2 \le r^2/t_{\min}^2$ pointwise, so $\lambda_{\max} \le t_{\min}^{-2}$ and $\lambda_{\min} \ge t_{\max}^{-2}$. In the example of (i), $\mathcal{L} = \mathbb{R}^2$ and the ratio $\frac{1}{2}(r_1^2 + r_2^2/\delta^2)/\frac{1}{2}(r_1^2 + r_2^2)$ runs from 1 to δ^{-2}, while $E(w_f)/E(w_r)$ at $c = b/\delta$ is at least $\delta^{-2}/4$. □

Proof of Proposition S1.3. $\widetilde{H} = H^{1/2}(I + B)H^{1/2}$ gives $\widetilde{H}^{-1} = H^{-1/2} Q H^{-1/2}$ and $Q = I - QB$. Hence $\widetilde{k}_x^\top \widetilde{H}^{-1}(Y - \widetilde{M}) - k_x^\top H^{-1}(Y - M) = \widetilde{g}^\top Q(W - U) - g^\top W = \widetilde{g}^\top W - \widetilde{g}^\top QBW - \widetilde{g}^\top QU - g^\top W = v^\top W - (Q\widetilde{g})^\top (BW + U)$, which is (i) since Q is symmetric; (ii) bounds its two terms. For (iii) write the same difference as $v^\top QW + g^\top(Q - I)W - \widetilde{g}^\top QU$ and use $\|Q\|_{\mathrm{op}} = \gamma_Q$ and $\|Q - I\|_{\mathrm{op}} = \|QB\|_{\mathrm{op}} = \eta_Q$. For (iv), $v = 0$, $U = 0$ and $B = sH^{-1}$ has eigenvalues $s/\lambda_i(H) \ge 0$, so $\eta_Q = \max_i (s/\lambda_i)/(1 + s/\lambda_i) = s/(\lambda_{\min}(H) + s)$. □

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Declaration of competing interest

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AI-assisted contributions

Claude Code (Anthropic) and Codex (OpenAI) implemented the data pipeline and the model comparisons, ran the experiments, produced the figures and tables from the run records, and drafted the manuscript. ChatGPT (OpenAI) developed the residual-error and inversion-stability arguments, checked the identities and wrote the conditioning diagnostics and their tests. The author supplied the data, set the questions, and reviewed the results, the proofs and the text.

Data and code availability

The code, the run records and this supplement are in the repository `https://github.com/yspennstate/emit-oco2-kernel-emulators`. The paper is part of the Math Brain project, whose public repository is `https://github.com/yspennstate/math-brain-public` [28].

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